

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2158

Firstenergy West Lorain 230 kV
September 30, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2158. The Interconnection Customer proposes a 40.8 MW solar photovoltaic facility interconnecting at the Firstenergy West Lorain 230 kV substation in the Ohio area. The request was received on August 14, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 437 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,289,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Aes / Knox 230kV
Contingency: Atascocita / North Cambridge Substation 230kV
Loading: 127.4% (Rating: 492 MVA)
DFAX: 18.2%
Pre-project loading: 125.9%
2. Facility: Marys Riv / Knox 230kV line
Contingency: Tidd Substation / Muskingum River 500kV line
Loading: 106.1% (Rating: 573 MVA)
DFAX: 7.6%
Pre-project loading: 105.6%
3. Facility: Tidd / Pelicano 138kV
Contingency: Lerdo Substation / Tidd 500kV
Loading: 101.2% (Rating: 312 MVA)
DFAX: 9.4%
Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Aes / Knox 230kV	Rebuild line with higher-capacity conductor	\$27,803,000	24% (\$6,732,000)
Knox / Marys Riv 230kV line	Reconductor line	\$15,408,000	12% (\$1,909,000)
Pelicano Substation / Tidd 138kV	Reconductor line	\$648,000	100% (\$648,000)
	Total allocated to project		\$9,289,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.